

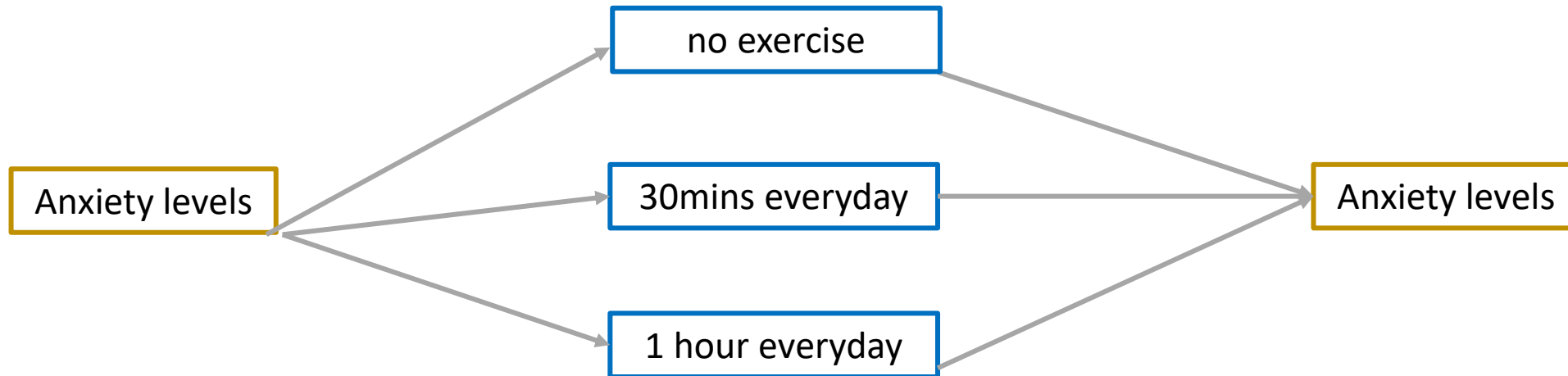
ANOVA

DV: Anxiety level

IV: Exercise

1 factor, 3 levels

IV – 3 levels



Can we perform multiple t-tests?

Why not just perform multiple t-tests?

To avoid Type I error – false positive

- For 'k' independent groups there are $k(k-1)/2$ possible t-tests
 - For 5 groups = $5(5-1)/2 = 10$ t-tests
 - For 4 groups = $4(4-1)/2 = 6$ t-tests
 - For 3 groups = $3(3-1)/2 = 3$ t-tests
- Using too many t-test comparisons increases the chances of finding random significant effects which may be due to chance. In reality there may be no difference between the groups/conditions

Risk of family-wise error – Increase Type I error

Bonferroni Correction

$$\frac{0.05 (\alpha)}{\text{No. of comparisons}} = 0.05/3 = 0.0167 \text{ (new } \alpha \text{ value)}$$

Can you do a t-test on this data?

Factor = Independent variable

2 Independent Variables - 2 levels each

Exercise – exercise vs control

Time of Day – morning vs evening

Two factorial design

Exercise-morning	Control-morning
Exercise-evening	Control-evening

2x2 factorial design

2 Independent Variables – different levels

Exercise – 30mins, 1 hour, 2 hours

Time of Day – morning vs evening

Two factorial design

30 mins-morning	1 hr-morning	2 hrs - morning
30 mins-evening	1 hr-evening	2 hrs - evening

3x2 factorial design

Can we perform multiple t-tests?

ANOVA

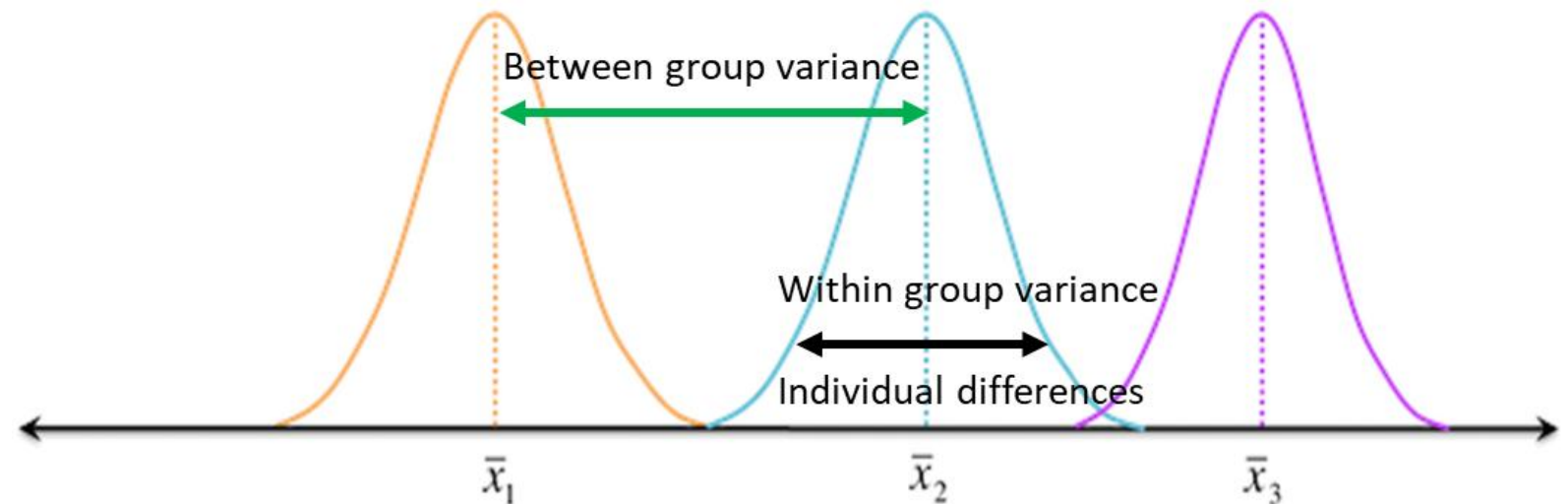
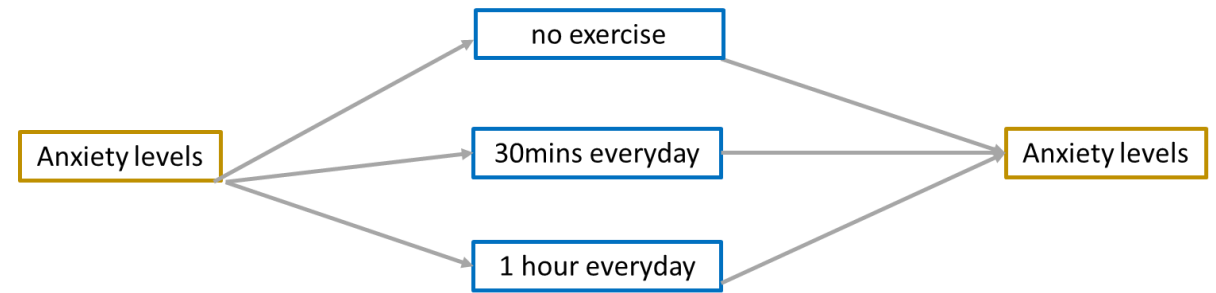
Analysis Of Variance

ANOVA performs all three comparisons simultaneously in one test.

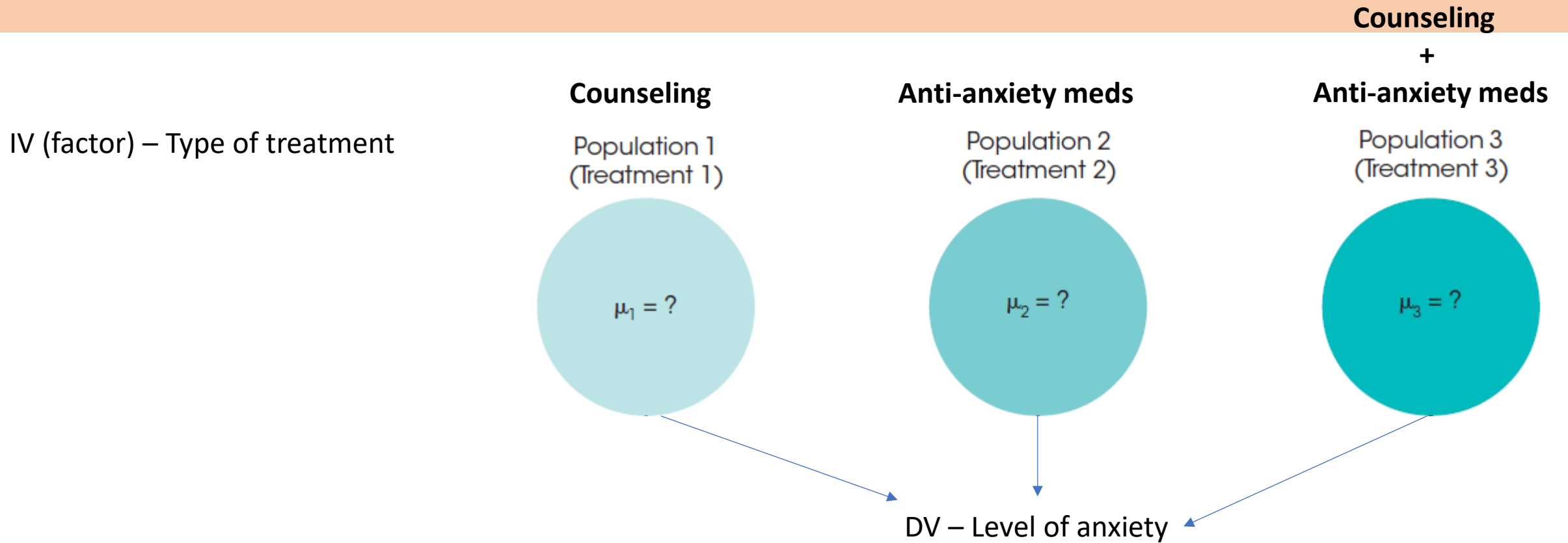
No matter how many different means are being compared, ANOVA uses one test with one alpha level to evaluate the difference in variance

$$F = \frac{\text{variance (differences) between sample means}}{\text{variance (differences) within sample}}$$

= difference due to treatment/experimental condition
individual differences in each condition



One Way ANOVA



H₀ - Null hypothesis – anxiety levels are equal across all groups after treatment (no difference between treatments)

H₁ - Alternate hypothesis ?

$$F = \frac{\text{variance between treatments}}{\text{variance within treatments}} = \frac{\text{systematic treatment effects} + \text{random, unsystematic differences}}{\text{random, unsystematic differences}} \geq 1 \text{ (treatment had an effect)}$$

$$F = \frac{0 + \text{random, unsystematic differences}}{\text{random, unsystematic differences}} \leq 1 \text{ (Treatment had no effect)}$$

	Source	SS (SS)	df	s ² (MS)	F
Effect of treatment	Between treatments	$\sum n_i (\bar{X}_i - \bar{\bar{X}})^2$	$k - 1$	$\frac{SS_b}{df_b} = \text{MSB}$	$F = \frac{\text{MSB}}{\text{MSW}}$
Random differences/error	Within treatments	$\sum (X_{ij} - \bar{X}_i)^2$	$N - k$	$\frac{SS_w}{df_w} = \text{MSW}$	
	Total	$\sum (X_{ij} - \bar{\bar{X}})^2$	$N - 1$		

X_{ij} = an individual observation

$\bar{X}_i$ = the mean of the i^{th} group

$\bar{\bar{X}}$ = the grand mean

k = the number of groups

n_i = the number of subjects in the i^{th} group

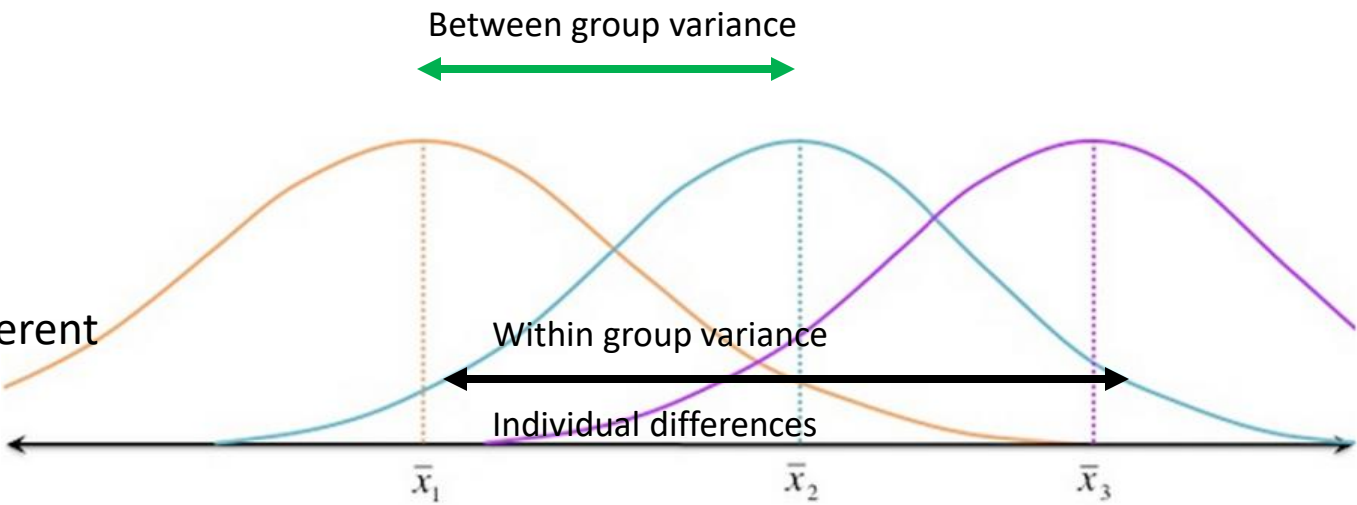
N = the number of subjects total

Source	SS	df	s ² (MS)	F
Between treatments	$\sum n_i(\bar{X}_i - \bar{\bar{X}})^2$	$k - 1$	$\frac{SS_b}{df_b}$	$\frac{s_b^2}{s_w^2}$
Within treatments	$\sum (X_{ij} - \bar{X}_i)^2$	$N - k$	$\frac{SS_w}{df_w}$	
Total	$\sum (X_{ij} - \bar{\bar{X}})^2$	$N - 1$		

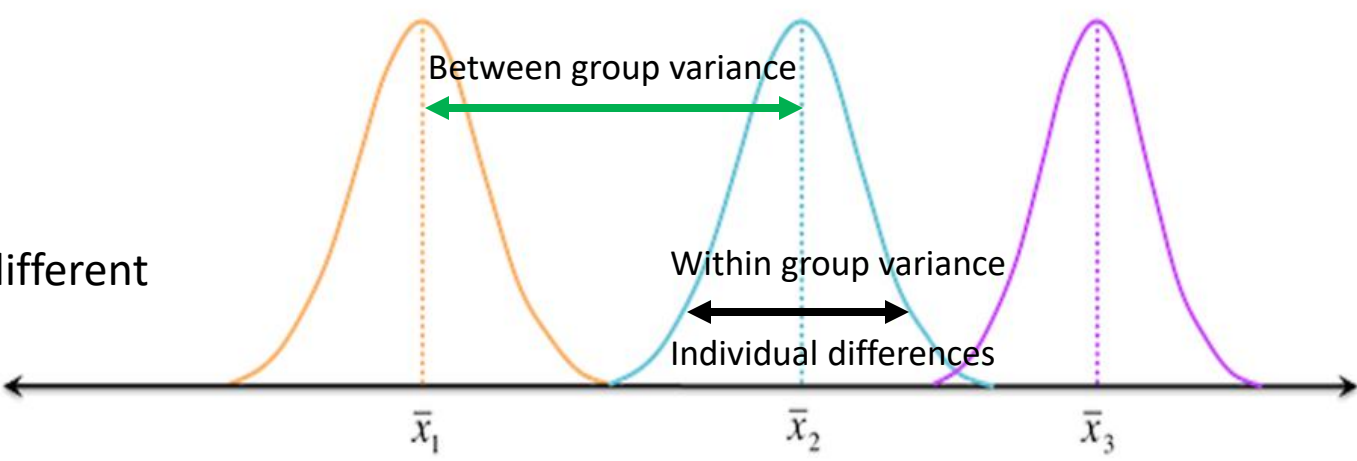
X_{ij} = an individual observation
 $\bar{X}_i$ = the mean of the i th group
 $\bar{\bar{X}}$ = the grand mean

k = the number of groups
 n_i = the number of subjects in the i th group
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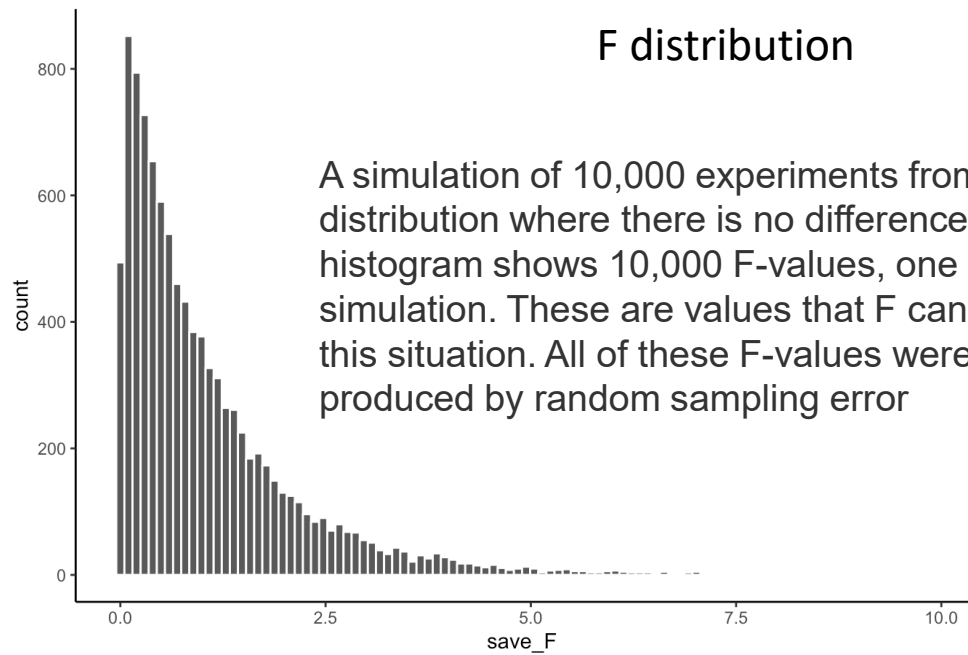
Group/Treatments are less likely to be significantly different



Groups/Treatments are more likely to be significantly different

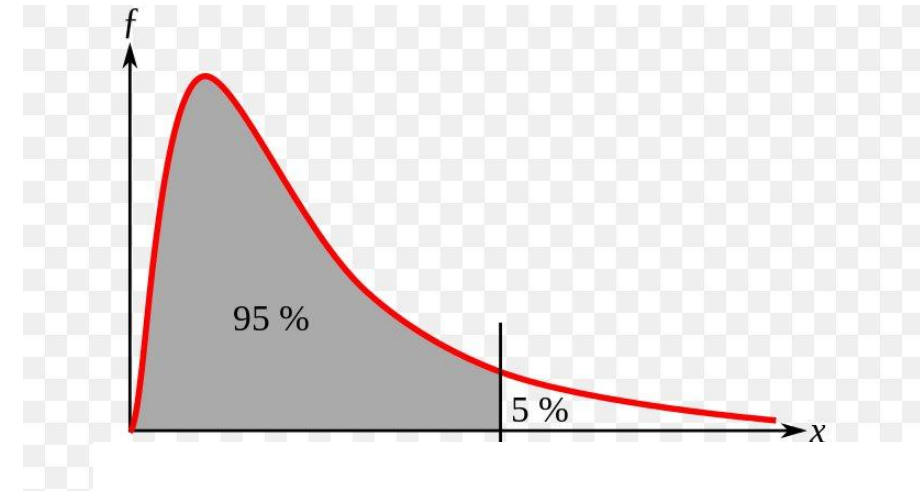


Simulated F-Distribution for Null



F distribution

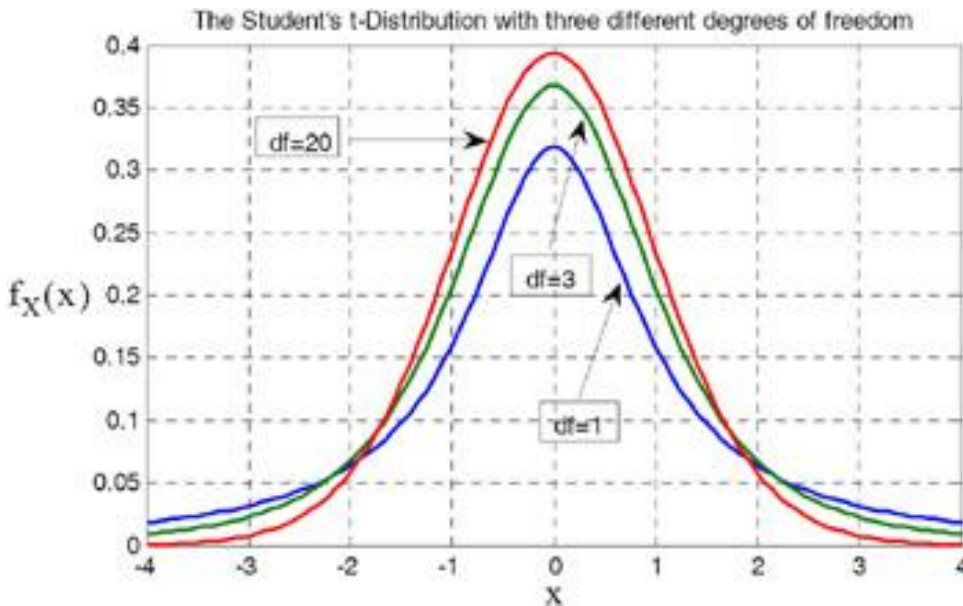
A simulation of 10,000 experiments from a null distribution where there is no differences. The histogram shows 10,000 F-values, one for each simulation. These are values that F can take in this situation. All of these F-values were produced by random sampling error



Why is F distribution positively skewed?

F distribution has only one tail – can only tell whether there is difference between the groups or not. Does not tell which group is better or worse.

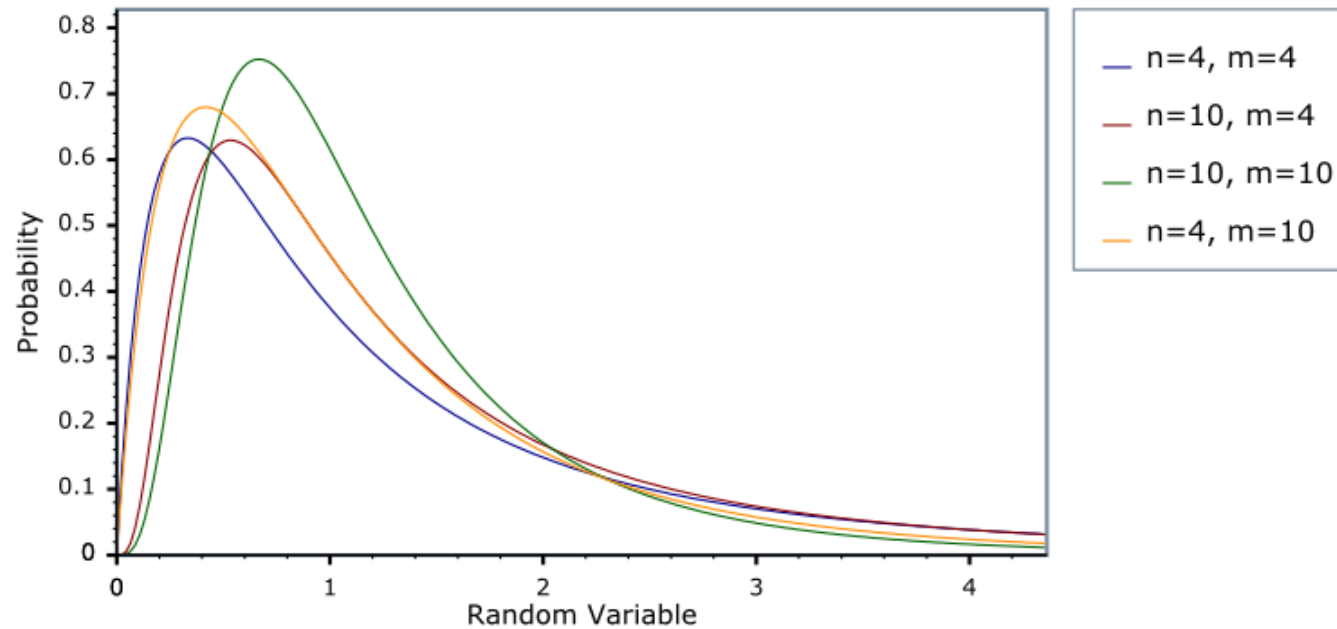
$$F = t^2$$



t distribution

(can be one tailed or two-tailed)

F Distribution PDF



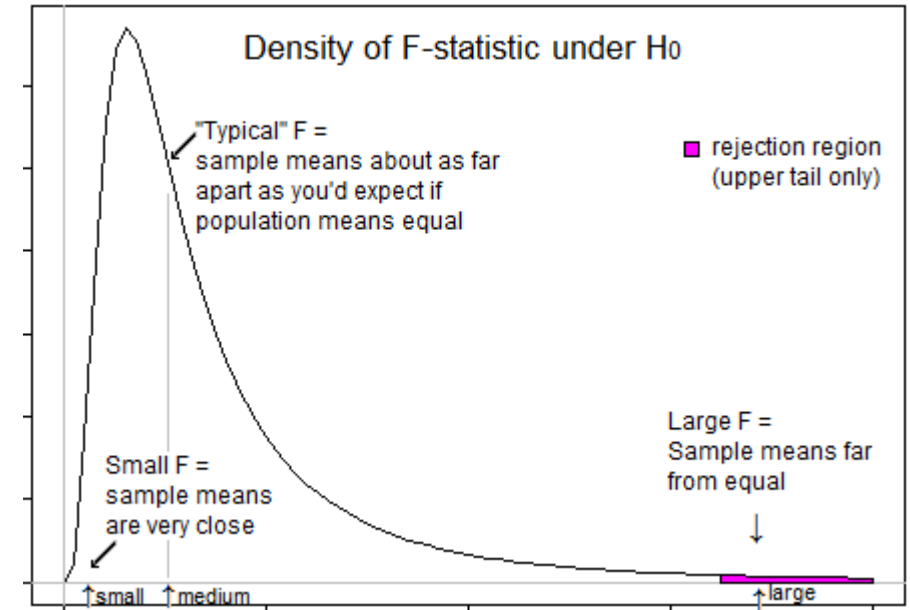
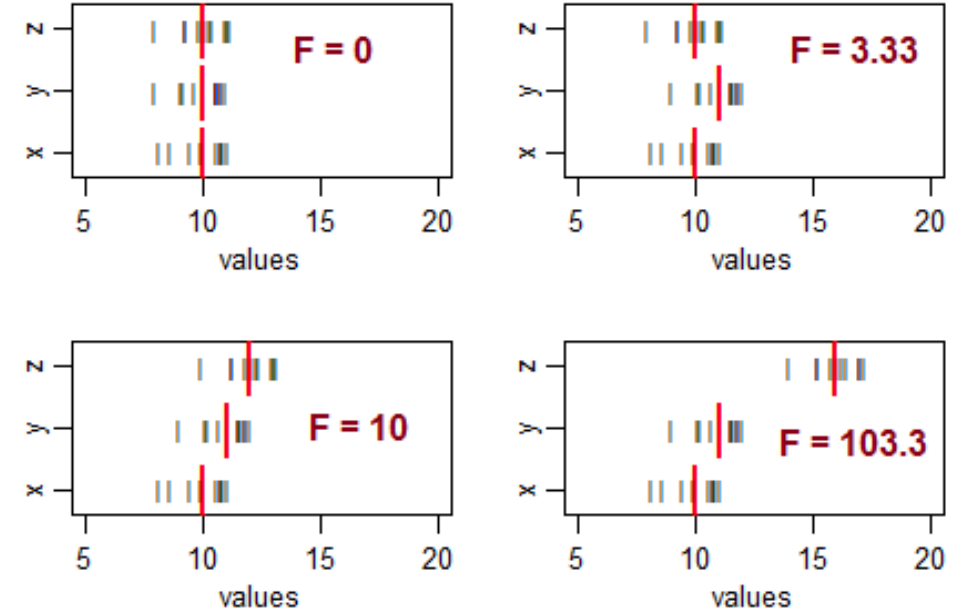
Groups	Count	Sum	Average	Variance
T1	3	33	11	81
T2	3	15	5	7
T3	3	15	5	27

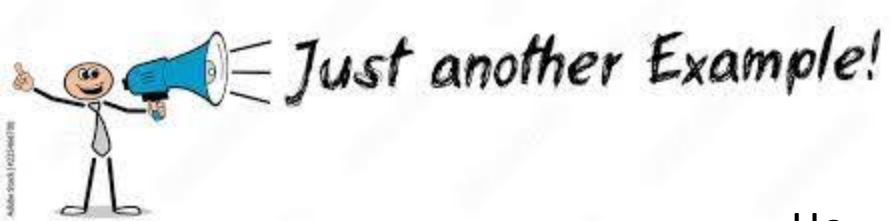
ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	72	2	36	0.93913	0.441736	5.143253
Within Groups	230	6	38.33333			
Total	302	8				

$$F = \frac{SSB}{SSW}$$

$F < 1$ – no effect

$F > 1$ – there might be an effect





One way (One factor, One IV) ANOVA

Ho – exam performance not affected by type of schooling

H1 – Type of schooling affects exam performance

FAKE DATA

Exam performance		
Home school	Boarding school	Regular Day school
89	85	91
75	78	88
49	59	84
87	77	81
84	63	91
68	88	75
88	71	69
78	73	93
77	69	95
93	80	85
67	72	87
79	68	84
69	66	83
88	59	80
91	70	77

Groups	Count	Sum	Average	Variance
Home school	15	1182	78.8	141.1714
Boarding school	15	1078	71.86667	73.98095
Regular Day school	15	1263	84.2	50.45714

ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	1146.711	2	573.3556	6.475922	0.003537	3.219942
Within Groups	3718.533	42	88.53651			
Total	4865.244	44				

$$F_{(2,42)} = 6.47, p < 0.01$$

or

$$F_{(2,42)} = 6.47, p = .003$$

Effect size for ANOVA

$$\eta^2 = \frac{SS_{Between}}{SS_{Total}} = \frac{1146.711}{4865.244} = 0.236$$

Eta-squared

$F_{(2,42)} = 6.47, p=.003, \eta^2 = .24$

Type of schooling explains 24% of variance in exam performance

We know there is difference between the groups, but which groups perform better or worse?

Table 1 Values of Effect Sizes and Their Interpretation

Kind of Effect Size	Small	Medium	Large
r	.10	.30	.50
d	0.20	0.50	0.80
η^2_p	.01	.06	.14
f^2	.02	.15	.35

Source: Cohen, J. (1992). A power primer. *Psychological Bulletin*, 112, 155–159. doi:10.1037/0033-2909.112.1.155

Why not just perform multiple t-tests?

To avoid Type I error – false positive

- For 'k' independent groups there are $k(k-1)/2$ possible t-tests
 - For 5 groups = $5(5-1)/2 = 10$ t-tests
 - For 4 groups = $4(4-1)/2 = 6$ t-tests
 - For 3 groups = $3(3-1)/2 = 3$ t-tests
- Using too many t-test comparisons increases the chances of finding random significant effects which may be due to chance. In reality there may be no difference between the groups/conditions

SOLUTION?

Hypothesis Driven
(like a one-tailed test)

Option 1 (Planned Contrasts): Pre-planned, therefore limited no. of comparisons.
You are not comparing all groups to one another, very specific comparisons
(so the risk of Type 1 error is controlled)

Exploratory
(like a two-tailed test)

Option 2 (Post Hoc tests) – All possible comparisons can be using special tests (to avoid Type I error).

Planned Comparison

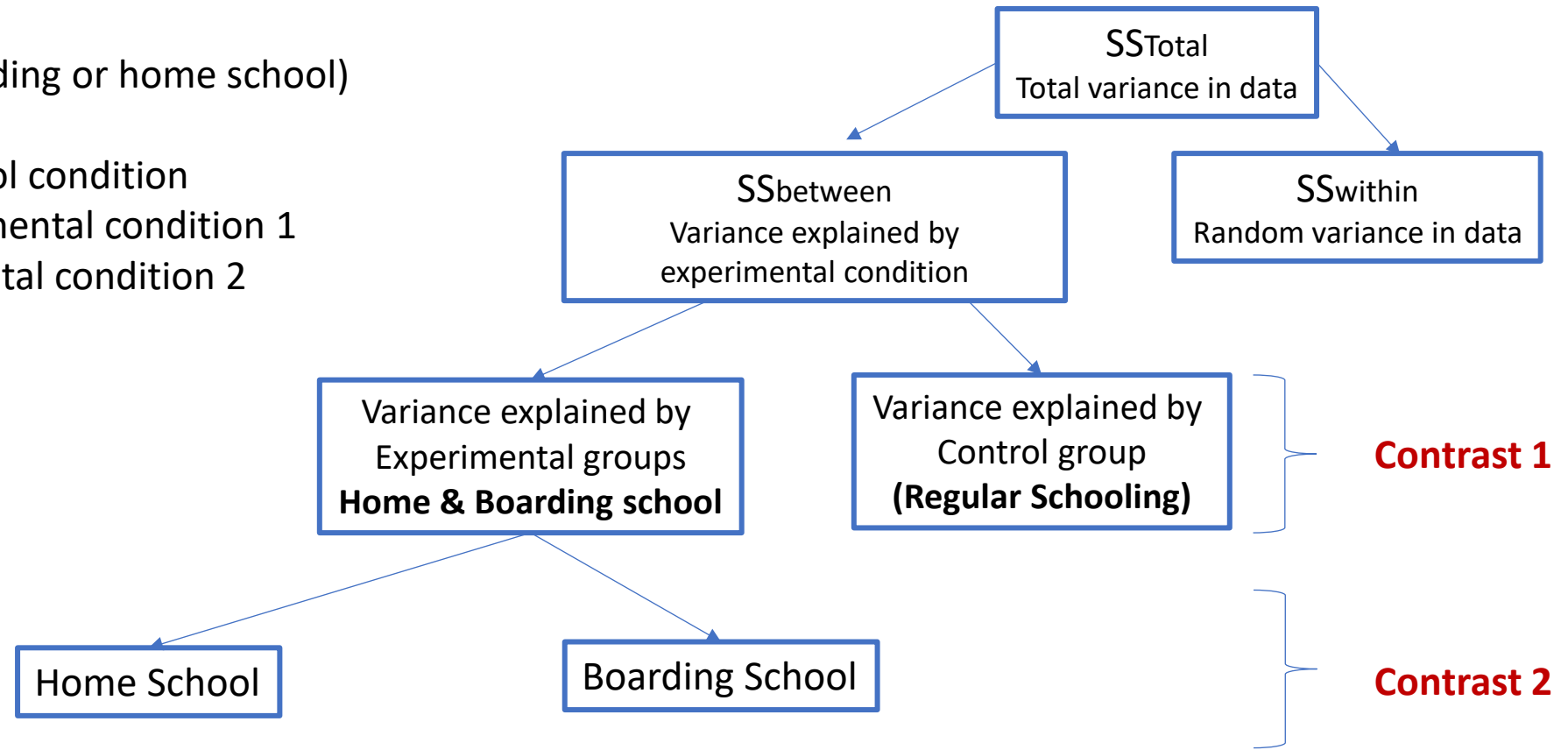
Planned comparison (contrast) – prior to experiment (based on the literature)

Regular schooling > (boarding or home school)

Regular schooling – control condition

Boarding school – Experimental condition 1

Home school – Experimental condition 2



But as the no. of planned comparisons increase (>2 comparisons), the alpha level has to adjusted/Bonferroni correction, again to avoid Type I error.

Bonferroni correction

Adjust the α level by the no. of comparisons

For 3 comparisons, $\alpha/3 = 0.05/3 = 0.0167 \sim 0.01$

Conduct 3 t-tests with $\alpha = 0.01$

Groupwise comparisons	T-test	Bonferroni correction corrected p value
Home vs Boarding	0.07780999	0.016666667
Boarding vs Regular	0.00019644	
Regular vs Home	0.14204177	

Good for small no. of comparisons, else risk of Type II error

Post-hoc test (Tukey's)

controlling Type I error rates (false positives) when conducting multiple pairwise comparisons.

Tukey's test requires that the sample size, n , be the same for all treatments.

Groups	Count	Sum	Average	Variance
Home school	15	1182	78.8	141.1714
Boarding school	15	1078	71.86667	73.98095
Regular Day school	15	1263	84.2	50.45714

ANOVA						
Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	1146.711	2	573.3556	6.475922	0.003537	3.219942
Within Groups	3718.533	42	88.53651			
Total	4865.244	44				

$$\text{Tukey's } HSD = q \sqrt{\frac{MS_{\text{within}}}{n}} = 3.44 \sqrt{(88.53/15)} = 6.95$$

q – studentized range statistic (depends on α, k, df)

- Quantifies the range of means relative to noise
- Adjusts for multiple comparisons
- Used to determine if the difference between any two means is statistically significant

The mean difference between any two samples must be more than 6.95 (at $\alpha=0.05$) to be significant.

$$M_{\text{Home}} - M_{\text{Regular}} = |78.8 - 84.2| = 5.4 \text{ (not significantly different)}$$

$$M_{\text{Boarding}} - M_{\text{Regular}} = |71.866 - 84.2| = 12.33 \text{ (significantly different)}$$

$$M_{\text{Home}} - M_{\text{Boarding}} = |78.8 - 71.87| = 6.93 \text{ (not significantly different)}$$

TABLE B.5 The Studentized Range Statistic (q)*

*The critical values for q corresponding to $\alpha = .05$ (lightface type) and $\alpha = .01$ (boldface type).

df for Error Term	k = Number of Treatments										
	2	3	4	5	6	7	8	9	10	11	12
5	3.64	4.60	5.22	5.67	6.03	6.33	6.58	6.80	6.99	7.17	7.32
	5.70	6.98	7.80	8.42	8.91	9.32	9.67	9.97	10.24	10.48	10.70
6	3.46	4.34	4.90	5.30	5.63	5.90	6.12	6.32	6.49	6.65	6.79
	5.24	6.33	7.03	7.56	7.97	8.32	8.61	8.87	9.10	9.30	9.48
7	3.34	4.16	4.68	5.06	5.36	5.61	5.82	6.00	6.16	6.30	6.43
	4.95	5.92	6.54	7.01	7.37	7.68	7.94	8.17	8.37	8.55	8.71
8	3.26	4.04	4.53	4.89	5.17	5.40	5.60	5.77	5.92	6.05	6.18
	4.75	5.64	6.20	6.62	6.96	7.24	7.47	7.68	7.86	8.03	8.18
9	3.20	3.95	4.41	4.76	5.02	5.24	5.43	5.59	5.74	5.87	5.98
	4.60	5.43	5.96	6.35	6.66	6.91	7.13	7.33	7.49	7.65	7.78
10	3.15	3.88	4.33	4.65	4.91	5.12	5.30	5.46	5.60	5.72	5.83
	4.48	5.27	5.77	6.14	6.43	6.67	6.87	7.05	7.21	7.36	7.49
11	3.11	3.82	4.26	4.57	4.82	5.03	5.20	5.35	5.49	5.61	5.71
	4.39	5.15	5.62	5.97	6.25	6.48	6.67	6.84	6.99	7.13	7.25
12	3.08	3.77	4.20	4.51	4.75	4.95	5.12	5.27	5.39	5.51	5.61
	4.32	5.05	5.50	5.84	6.10	6.32	6.51	6.67	6.81	6.94	7.06
13	3.06	3.73	4.15	4.45	4.69	4.88	5.05	5.19	5.32	5.43	5.53
	4.26	4.96	5.40	5.73	5.98	6.19	6.37	6.53	6.67	6.79	6.90
14	3.03	3.70	4.11	4.41	4.64	4.83	4.99	5.13	5.25	5.36	5.46
	4.21	4.89	5.32	5.63	5.88	6.08	6.26	6.41	6.54	6.66	6.77
15	3.01	3.67	4.08	4.37	4.59	4.78	4.94	5.08	5.20	5.31	5.40
	4.17	4.84	5.25	5.56	5.80	5.99	6.16	6.31	6.44	6.55	6.66
16	3.00	3.65	4.05	4.33	4.56	4.74	4.90	5.03	5.15	5.26	5.35
	4.13	4.79	5.19	5.49	5.72	5.92	6.08	6.22	6.35	6.46	6.56
17	2.98	3.63	4.02	4.30	4.52	4.70	4.86	4.99	5.11	5.21	5.31
	4.10	4.74	5.14	5.43	5.66	5.85	6.01	6.15	6.27	6.38	6.48
18	2.97	3.61	4.00	4.28	4.49	4.67	4.82	4.96	5.07	5.17	5.27
	4.07	4.70	5.09	5.38	5.60	5.79	5.94	6.08	6.20	6.31	6.41
19	2.96	3.59	3.98	4.25	4.47	4.65	4.79	4.92	5.04	5.14	5.23
	4.05	4.67	5.05	5.33	5.55	5.73	5.89	6.02	6.14	6.25	6.34
20	2.95	3.58	3.96	4.23	4.45	4.62	4.77	4.90	5.01	5.11	5.20
	4.02	4.64	5.02	5.29	5.51	5.69	5.84	5.97	6.09	6.19	6.28
24	2.92	3.53	3.90	4.17	4.37	4.54	4.68	4.81	4.92	5.01	5.10
	3.96	4.55	4.91	5.17	5.37	5.54	5.69	5.81	5.92	6.02	6.11
30	2.89	3.49	3.85	4.10	4.30	4.46	4.60	4.72	4.82	4.92	5.00
	3.89	4.45	4.80	5.05	5.24	5.40	5.54	5.65	5.76	5.85	5.93
40	2.86	3.44	3.79	4.04	4.23	4.39	4.52	4.63	4.73	4.82	4.90
	3.82	4.37	4.70	4.93	5.11	5.26	5.39	5.50	5.60	5.69	5.76
60	2.83	3.40	3.74	3.98	4.16	4.31	4.44	4.55	4.65	4.73	4.81
	3.76	4.28	4.59	4.82	4.99	5.13	5.25	5.36	5.45	5.53	5.60
120	2.80	3.36	3.68	3.92	4.10	4.24	4.36	4.47	4.56	4.64	4.71
	3.70	4.20	4.50	4.71	4.87	5.01	5.12	5.21	5.30	5.37	5.44
∞	2.77	3.31	3.63	3.86	4.03	4.17	4.28	4.39	4.47	4.55	4.62
	3.64	4.12	4.40	4.60	4.76	4.88	4.99	5.08	5.16	5.23	5.29

Table 29 of E. Pearson and H. O. Hartley, *Biometrika Tables for Statisticians*, 2nd ed. New York: Cambridge University Press, 1958. Adapted and reprinted with permission of the Biometrika trustees.

Other post-hoc tests

Games-Howell Test

For unequal variance (unequal sample size)

Calculations are the same as Tukey's but df is calculated using the formula used for unequal sample t-test (Slide 1)

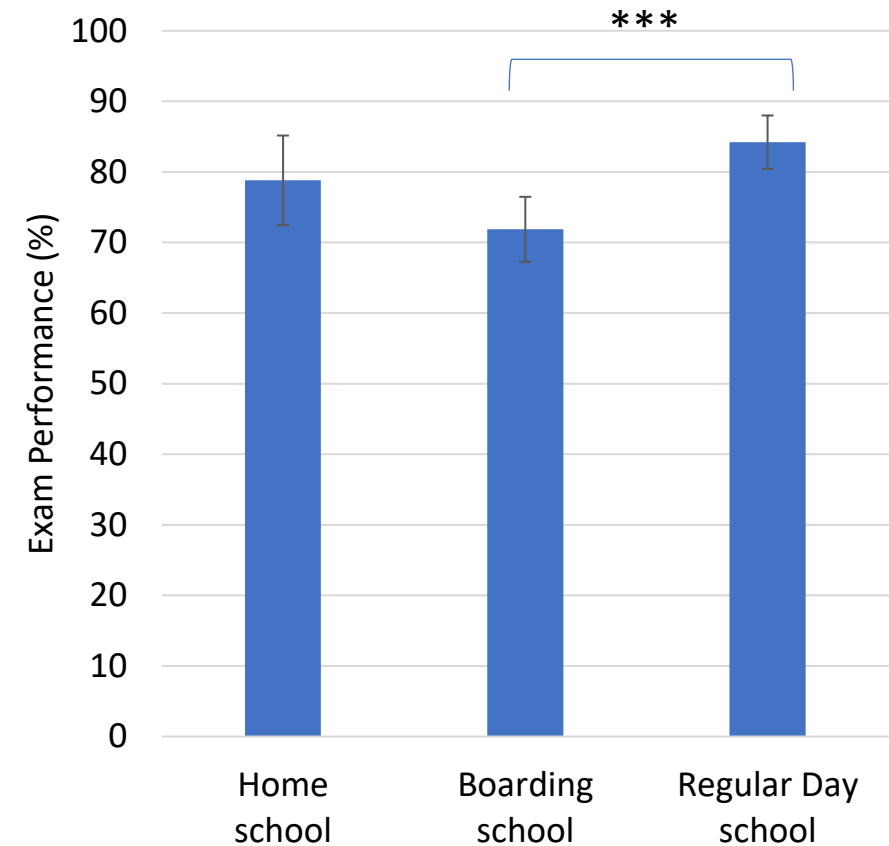
For unequal sample size

$$\text{degrees of freedom, df} = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1} \left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1} \left(\frac{s_2^2}{n_2}\right)^2}$$

Reporting results

- Using a one way ANOVA we observed that the schooling method has a significant effect on exam performance $F_{(2,42)} = 6.47$, $p = .003$, $\eta^2 = .24$
- Using Bonferroni post-hoc test, we found that regular school resulted in better exam performance than boarding school ($p < .001$). There was no significant difference between the other groups.

* $p < 0.05$
** $p < 0.01$
*** $p < 0.001$



Error bars denote confidence intervals

ANOVA assumptions

1. The populations from which the samples are selected must be normal (**parametric vs non-parametric**)

– **Shapiro-Wilk test / Kolmogorov-Smirnov test**

If violated – use Kruskal –Wallis Test

Typically for $n > 25$ in each group, normality can be overlooked in ANOVA

2. The populations from which the samples are selected must have equal variances* (**homogeneity of variance**).

– **Levene's or Hartley's F -max test for homogeneity of variance**

If violated, solution -

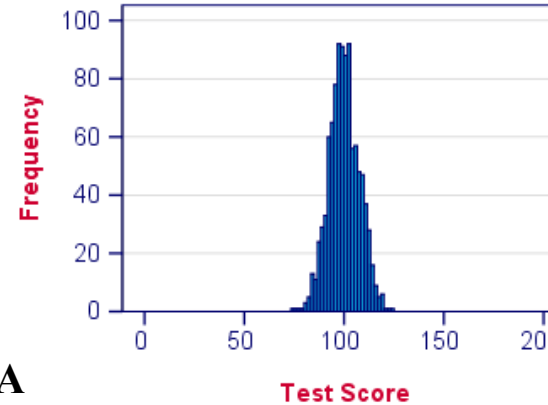
1. Collect more samples and equate samples in all groups.

2. Data transformations - natural log or square root transformations

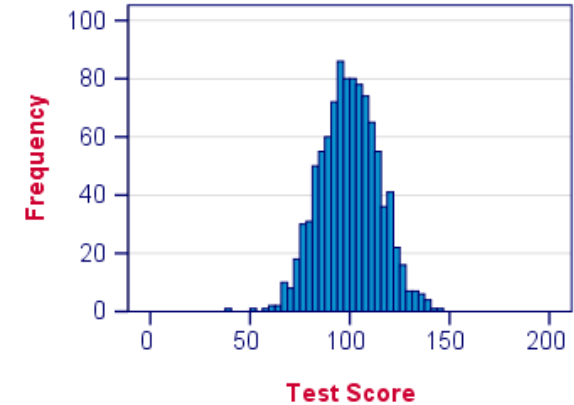
Consequences -- once you transform a variable and conduct your analysis, you can only interpret the transformed variable. You cannot provide an interpretation of the results based on the untransformed variable values.

NORMAL DISTRIBUTIONS WITH SIMILAR MEANS, DIFFERENT VARIANCES.

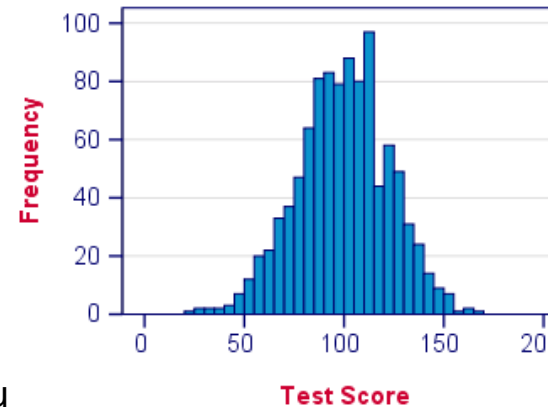
Histogram. Mean = 100 | Variance = 25



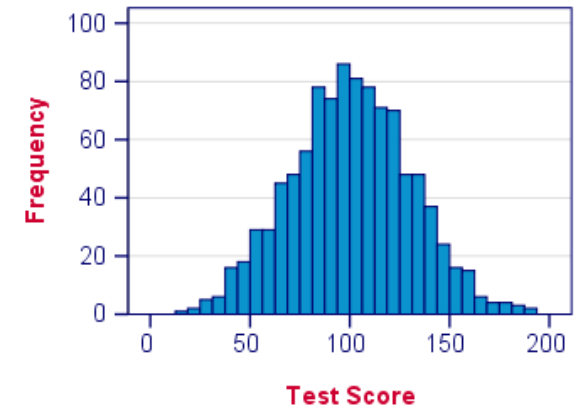
Histogram. Mean = 100 | Variance = 100



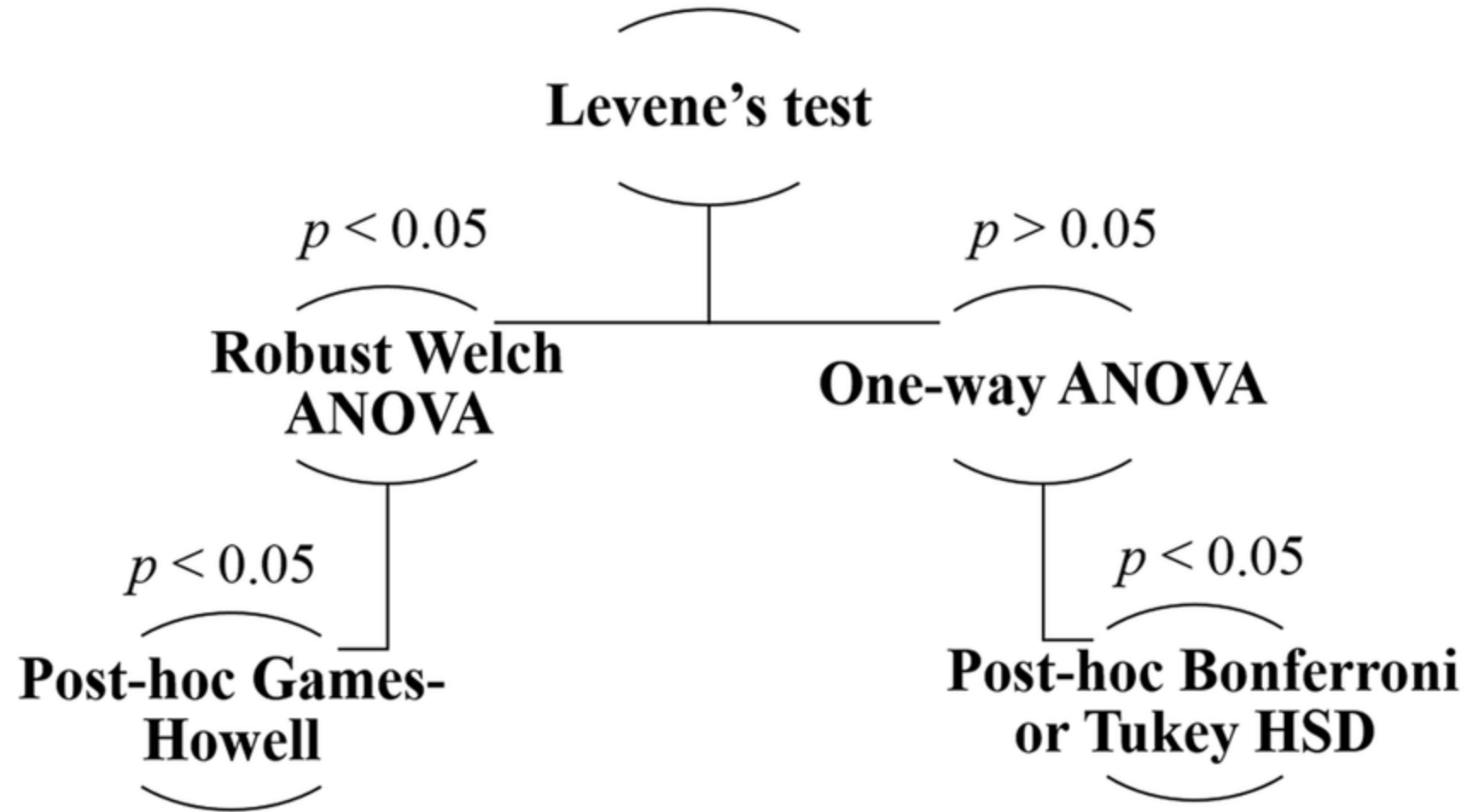
Histogram. Mean = 100 | Variance = 225



Histogram. Mean = 100 | Variance = 400

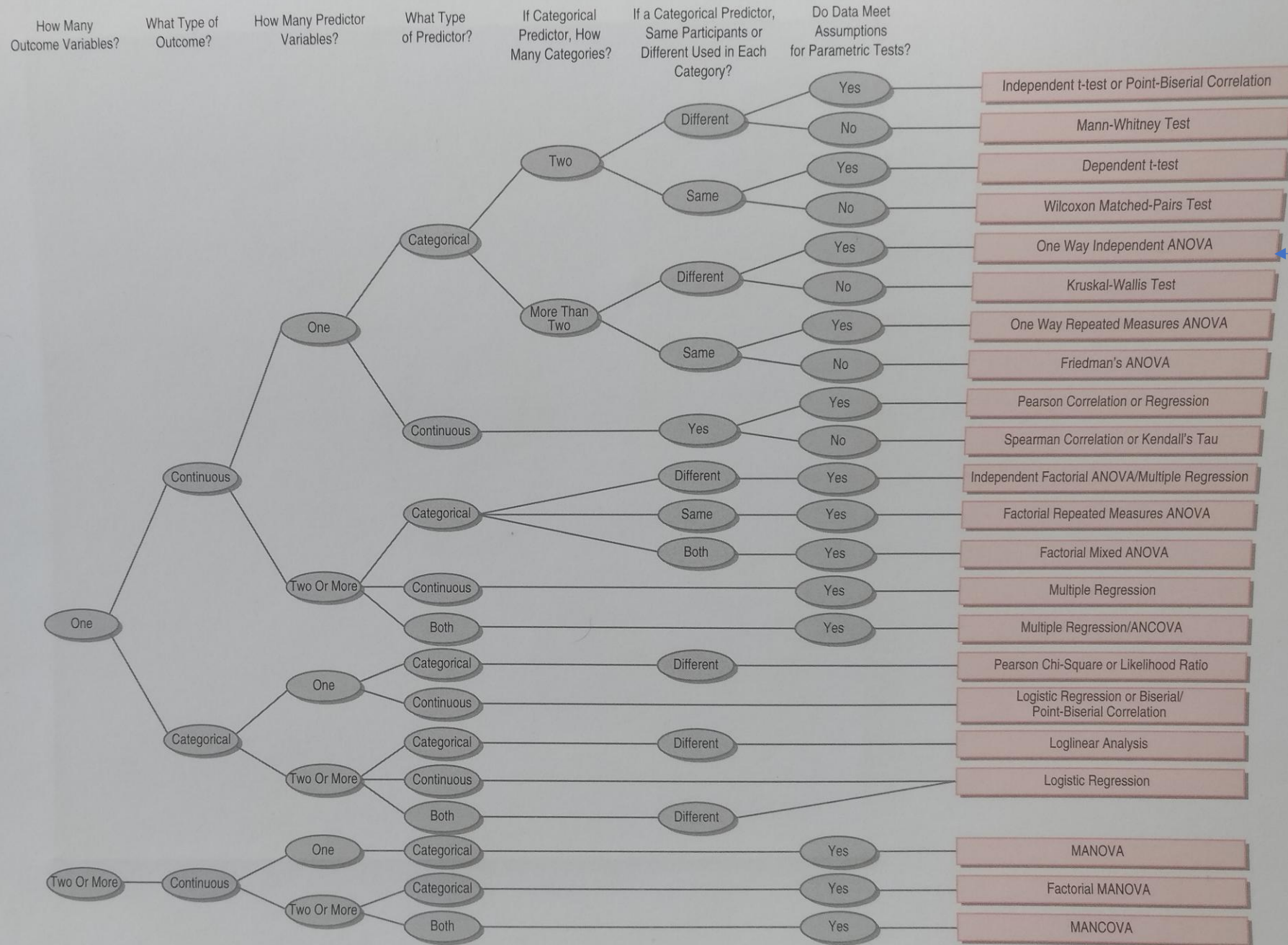


ANOVA is a robust statistical test, slight violations of assumptions has minor effects on the test outcomes. As long as the largest variance < 4-5 times smallest variance, ANOVA results are valid



Steps in ANOVA

- Check for normality (parametric vs non-parametric tests)
- Check homogeneity of variances (spread of the data is equal across groups or not)
- Choose the appropriate test (ANOVA, Kruskal-Wallis, Welch)
- Only if main effect (F) significant, use a post-hoc test
- Report effect size for significant effects
- Plot analyzed data

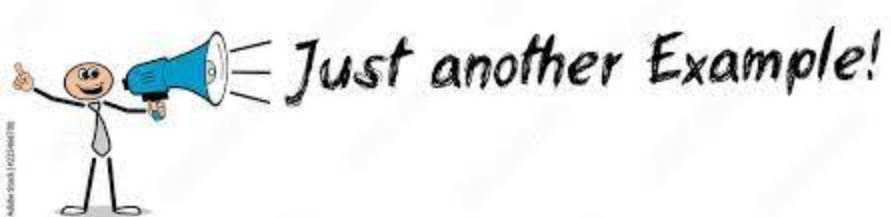


Welch ANOVA Test

no

yes

homogenous



One way (One factor, One IV) ANOVA

Testing homogeneity

Ho – variance across groups is equal

H1 – variance across groups is unequal

Exam performance		
Home school	Boarding school	Regular Day school
89	85	91
75	78	88
49	59	84
87	77	81
84	63	91
68	88	75
88	71	69
78	73	93
77	69	95
93	80	85
67	72	87
79	68	84
69	66	83
88	59	80
91	70	77

Tests of Homogeneity of Variances

		Levene Statistic	df1	df2	Sig.
exam_performance	Based on Mean	1.675	2	42	.200
	Based on Median	1.648	2	42	.205
	Based on Median and with adjusted df	1.648	2	35.806	.207
	Based on trimmed mean	1.690	2	42	.197

Tests of Normality

		Kolmogorov-Smirnov ^a			<50 samples Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
exam_performance	school_code						
	home school	.155	15	.200*	.905	15	.115
	boarding school	.114	15	.200*	.969	15	.847
	regular scchool	.100	15	.200*	.975	15	.923

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

One way Repeated Measures ANOVA

Advantage?

Disadvantage?

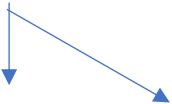
Experimental
Design is equally
important

subjects	counseling	Anti-anxiety meds	both
1	20	11	2
2	6	2	7
3	2	11	2

subjects	T1 counseling	T2 counselling	T3 counselling
1	20	11	2
2	6	2	7
3	2	11	2

One way Repeated Measures ANOVA

$$SS_{total} = SS_{between} + SS_{within}$$



$$SS_{total} = SS_{between} + SS_{subjects} + SS_{error}$$

	Dependent variable (DV)		
Participants	Timepoint 1	Timepoint 2	Timepoint 3
1	20	11	2
2	6	2	7
3	2	11	2

Source of variance	SS	df	MS	F
Between	$SS_{between}$	k-1	$MS_{between} = \frac{SS_{between}}{k-1}$	$F = \frac{MS_{between}}{MS_{error}}$
Within	SS_{within}			
	$SS_{subjects}$	n-1		
	$SS_{error} = SS_{within} - SS_{subjects}$	(k-1)(n-1)	$MS_{error} = \frac{SS_{error}}{(k-1)(n-1)}$	
Total	SS_{total}	N-1		

Mauchly's sphericity test

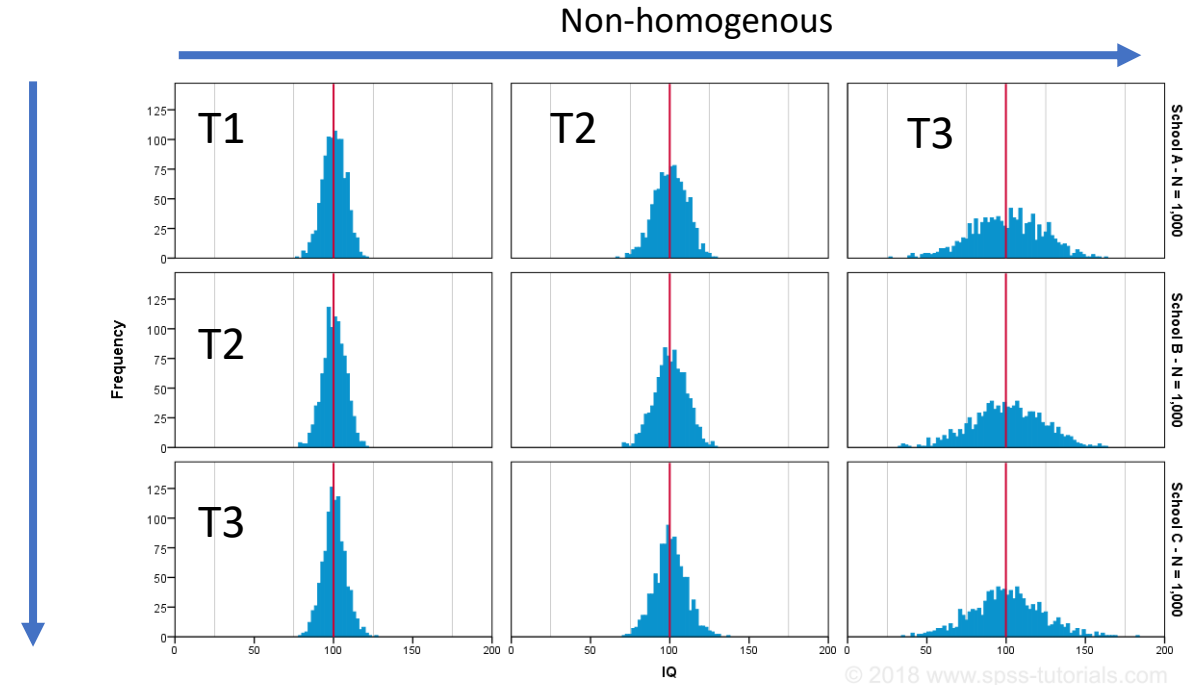
Sphericity → condition where the variances of related groups (levels T1, T2 , T3) are equal.

- Analogous to homogeneity of variances
- Used specifically in repeated measures testing

$$Df_{\text{between OR } df_{\text{time/condition}}} = (k - 1)$$

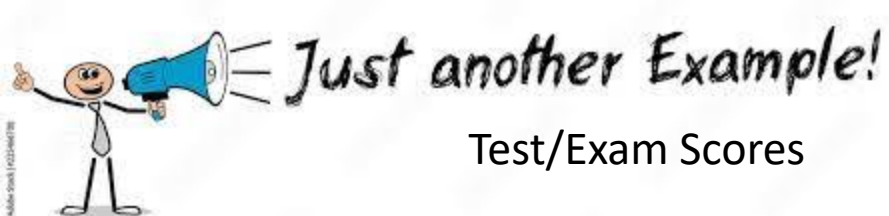
$$df_{\text{error}} = (k - 1)(n - 1)$$

homogenous



Steps in ANOVA

- Check for normality
- Check homogeneity of variances (different participants in different groups, Factorial ANOVA)
- Check for sphericity of variances (same participants across groups, repeated measures ANOVA)
- Choose the appropriate test
- Only if main effect (F) significant, use a post-hoc test
- Report effect size for significant effects
- Plot analyzed data



One way (One DV) repeated measures ANOVA

Test/Exam Scores

Student	Reread	Answer Prepared Questions	Create and Answer Questions
A	2	5	8
B	3	9	6
C	8	10	12
D	6	13	11
E	5	8	11
F	6	9	12

Mauchly's Test of Sphericity^a

Measure: test_score

Within Subjects Effect	Mauchly's W	Approx. Chi-Square	df	Sig.	Greenhouse-Geisser	Epsilon ^b Huynh-Feldt
factor1	.372	3.957	2	.138	.614	.712

Tests the null hypothesis that the error covariance matrix of the orthonormalized transformed dependent variables is proportional to an identity matrix.

a. Design: Intercept
Within Subjects Design: factor1

b. May be used to adjust the degrees of freedom for the averaged tests of significance. Corrected tests are displayed in the Tests of Within-Subjects Effects table.

Tests of Within-Subjects Effects

Measure: test_score

Source		Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
factor1	Sphericity Assumed	84.000	2	42.000	19.091	<.001	.792
	Greenhouse-Geisser	84.000	1.228	68.380	19.091	.004	.792
	Huynh-Feldt	84.000	1.424	58.991	19.091	.002	.792
	Lower-bound	84.000	1.000	84.000	19.091	.007	.792
Error(factor1)	Sphericity Assumed	22.000	10	2.200			
	Greenhouse-Geisser	22.000	6.142	3.582			
	Huynh-Feldt	22.000	7.120	3.090			
	Lower-bound	22.000	5.000	4.400			

Tests of Normality

<50 samples

	Kolmogorov-Smirnov ^a			Shapiro-Wilk		
	Statistic	df	Sig.	Statistic	df	Sig.
reread	.176	6	.200*	.955	6	.783
answer_questions	.184	6	.200*	.957	6	.799
create_answer_questions	.325	6	.047	.827	6	.101

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

Using a one way repeated measures ANOVA we observed that strategy for studying in preparation for a test had an effect on exam score $F_{(2,10)} = 19.09$, $p < .001$, $\eta^2 = .79$

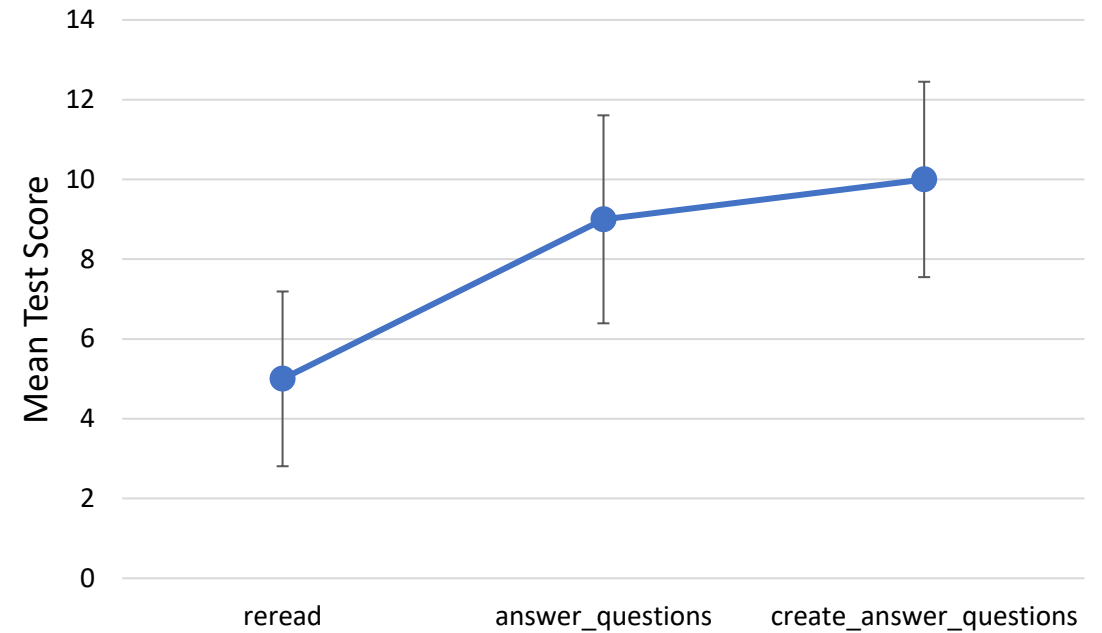
Tests of Within-Subjects Contrasts

Measure: test_score

Source	factor1	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared
factor1	Linear	75.000	1	75.000	93.750	<.001	.949
	Quadratic	9.000	1	9.000	2.500	.175	.333
Error(factor1)	Linear	4.000	5	.800			
	Quadratic	18.000	5	3.600			

Using a one way repeated measures ANOVA we observed that strategy for studying in preparation for a test had an effect on exam score $F_{(2,10)} = 19.09$, $p < .001$, $\eta^2 = .79$

Within-subjects contrasts revealed that there was a linear trend $F_{(1,5)} = 93.75$, $p < .001$, $\eta^2 = .95$



Error bars denote standard deviations

Friedman's Test (non-normal repeated measures)

$$M = \frac{12}{Nk(k+1)} \sum R_i^2 - 3N(k+1)$$

Where, k = number of columns (treatments)

n = number of rows (blocks)

R_j = sum of the ranks

Related-Samples Friedman's Two-Way Analysis of Variance by Ranks Summary

Total N	6
Test Statistic	9.333
Degree Of Freedom	2
Asymptotic Sig.(2-sided test)	.009

Student	Reread	Answer Prepared Questions	Create and Answer Questions
A	2	5	8
B	3	9	6
C	8	10	12
D	6	13	11
E	5	8	11
F	6	9	12

Rank		
1	2	3
1	3	2
1	2	3
1	3	2
1	2	3
1	2	3
Sum = 6	14	16

IV – categorical

DV – continuous (interval, ratio)

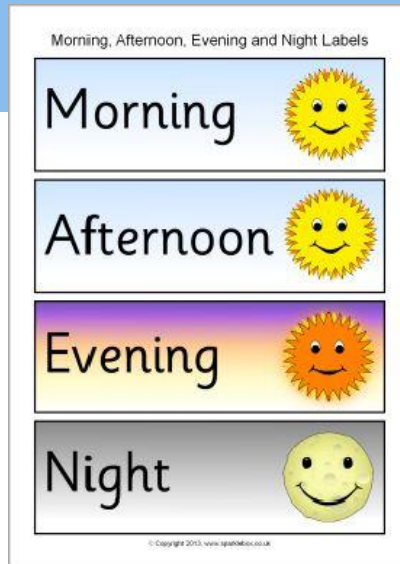
	Levene's Test	Independent factor 1 IV > 2 groups	Dependent (Related) Samples 1 DV > 2 timepoints
Parametric (normal)	Homogeneous	One way ANOVA	Repeated measures ANOVA
	Non homogenous	Welch ANOVA	Sphericity correction (report
Non-parametric		<i>Kruskal-Wallis ANOVA</i>	Friedman's ANOVA

Does time of day affect reaction time?

Independent Variables

(Factor)

Time of day



Dependent Variables

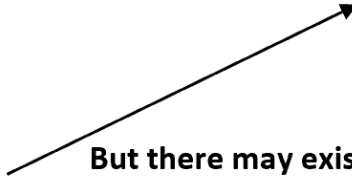
(Response)

Reaction time

Which test?

One way ANOVA

We're interested in the relationship between one or more explanatory variables and a response variable.



But there may exist some other variable (a *covariate*) that also affects the response variable.

Explanatory Variable(s)

Response Variable

Covariate

Analysis of Covariance (ANCOVA)

Independent Variables

(Factor)

Time of day

(Covariate)

Sleep
(hours slept the
night before)

Dependent Variable

(Response)

Reaction time



Advantages of ANCOVA

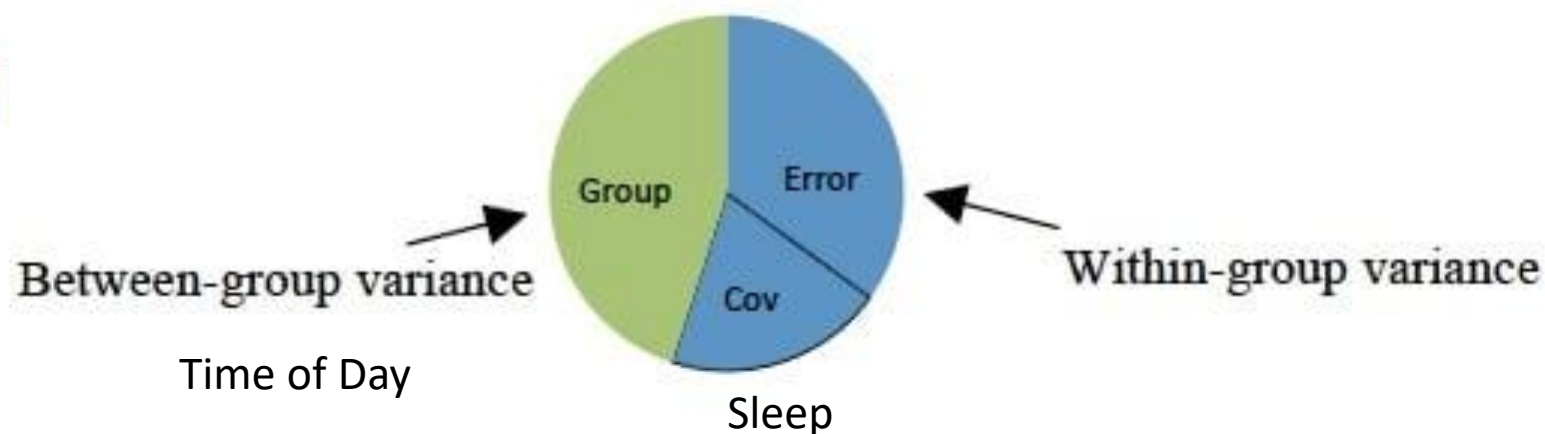
- **Reduces Error Variance**

- By explaining some of the unexplained variance (SS_R) the error variance in the model can be reduced.

- **Greater Experimental Control:**

- By holding known extraneous variables constant, we gain greater insight into the effect of the predictor variable(s).

ANDY FIELD

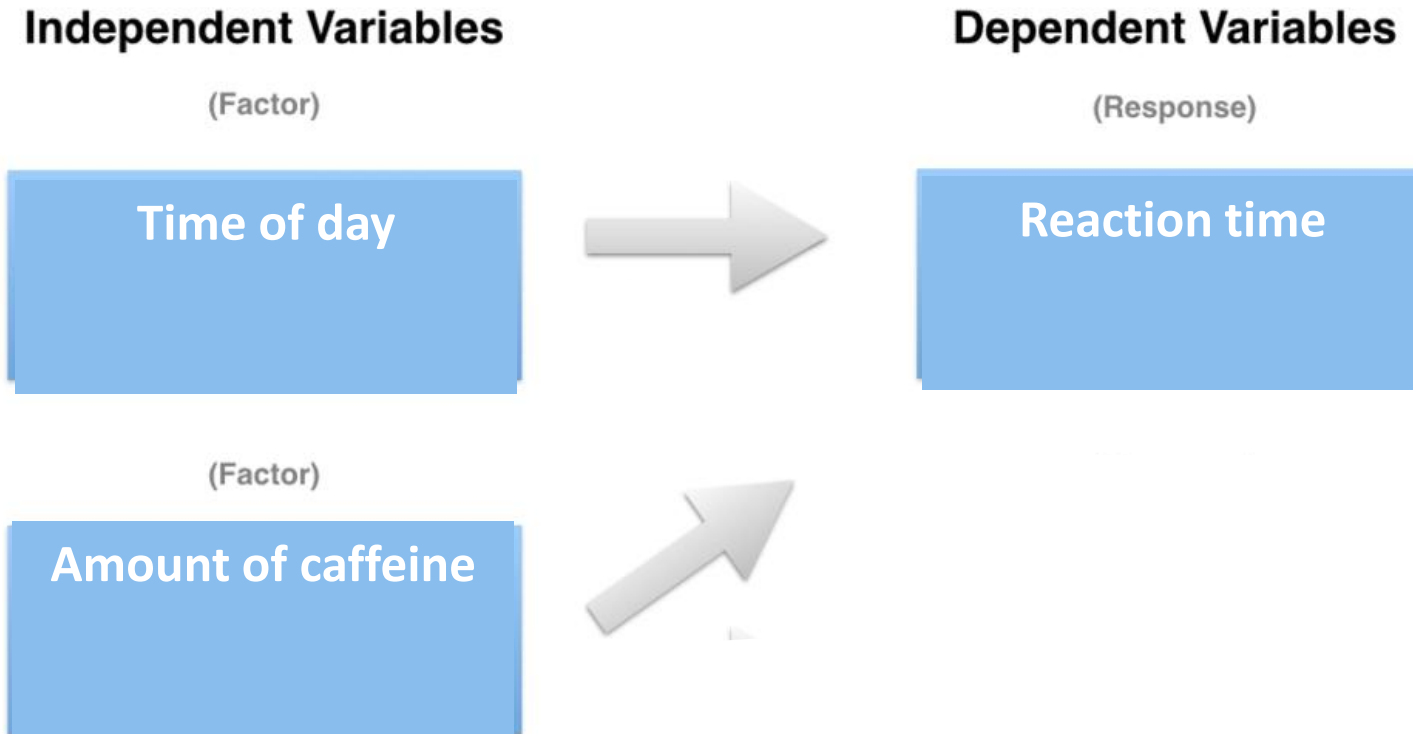


Do people with private health insurance visit their physicians more frequently than people with no insurance or other types of insurance?

Covariate?

Factorial (Univariate) ANOVA

≥ 2 IVs, 1 DV



Factorial ANOVA

≥ 2 IVs, 1 DV

2x2 Design		Time of Day	
		Morning	Afternoon
Caffeine	Some Caffeine	Reaction time	Reaction time
	No Caffeine	Reaction time	Reaction time

IV – categorical

DV – continuous (interval, ratio)

2 Independent variables, 1 Dependent variable

2 x 2 ANOVA

2x3 Design		Time of Day	
		Morning	Afternoon
Caffeine	1 coffee	Reaction time	Reaction time
	2 coffees	Reaction time	Reaction time
	3 coffees	Reaction time	Reaction time

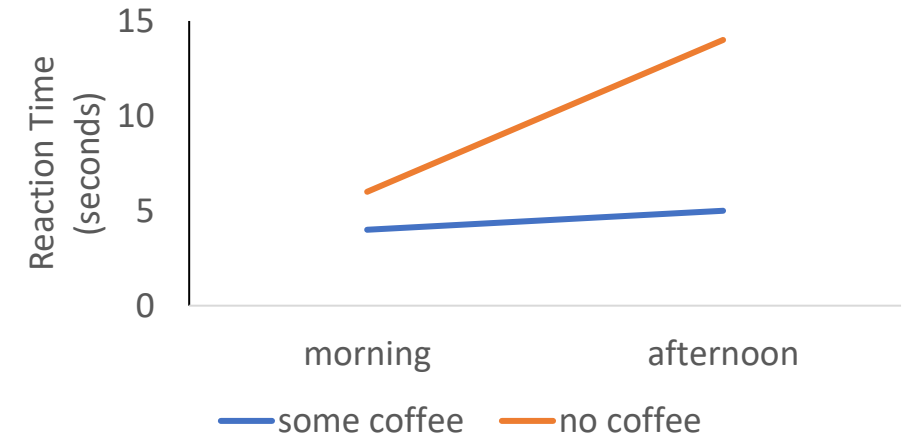
2 Independent variables, 1 Dependent variable

3 x 2 ANOVA

Factorial ANOVA

Interaction between time of day and amount of caffeine intake

2x2 Design		Time of Day	
		Morning	Afternoon
Caffeine	Some Caffeine	Reaction time	Reaction time
	No Caffeine	Reaction time	Reaction time



Source	SS	df	S^2	F
(Between) Row	$\sum [N_{\text{row}}(M_{\text{row}} - M_o)^2]$	rows-1	$\frac{SS_r}{df_r}$	$\frac{S^2_r}{S^2_w}$
(Between) Column	$\sum [N_{\text{col}}(M_{\text{col}} - M_o)^2]$	columns-1	$\frac{SS_c}{df_c}$	$\frac{S^2_c}{S^2_w}$
(Between) Interaction	$\sum [N_{\text{cell}}(M_{\text{cell}} - M_o)^2] - SS_{\text{row}} - SS_{\text{col}}$	(rows-1)(columns-1)	$\frac{SS_i}{df_i}$	$\frac{S^2_i}{S^2_w}$
Within	$\sum (X - M_{\text{cell}})^2$	N - cells	$\frac{SS_w}{df_w}$	X: individual score N: number of scores M_o : overall mean
Total	$\sum (X - M_o)^2$	N - 1		

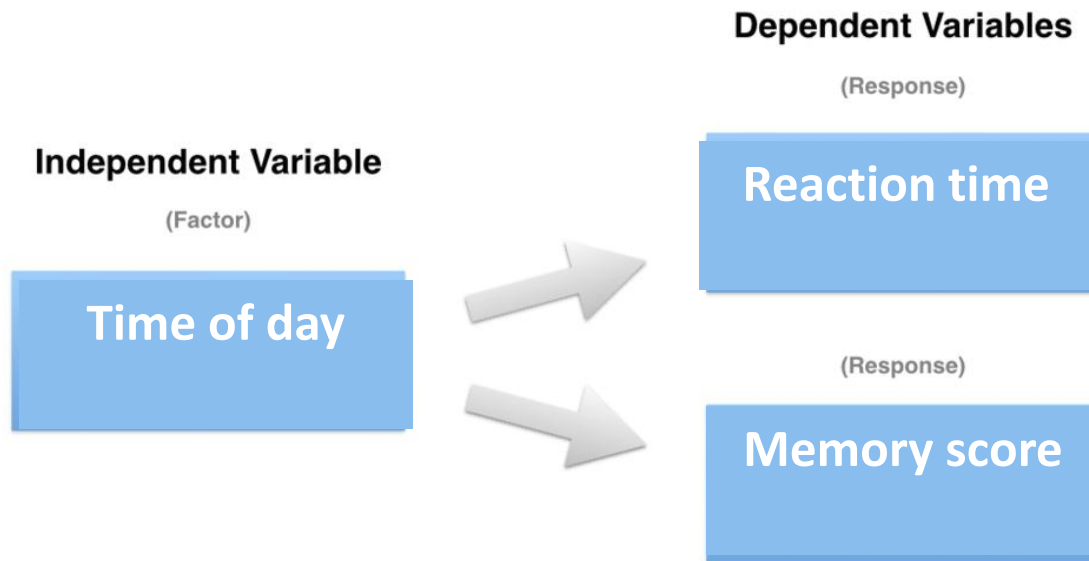
Source	SS	df	S ²	F																
(Between) Row	$\sum [N_{\text{row}}(M_{\text{row}} - M_o)^2]$	rows- 1	$\frac{SS_r}{df_r}$	$\frac{S^2_r}{S^2_w}$																
(Between) Column	$\sum [N_{\text{col}}(M_{\text{col}} - M_o)^2]$	columns-1	$\frac{SS_c}{df_c}$	$\frac{S^2_c}{S^2_w}$																
(Between) Inter-action	$\sum [N_{\text{cell}}(M_{\text{cell}} - M_o)^2]$ - SS _{row} - SS _{col}	(rows- 1) (columns-1)	$\frac{SS_i}{df_i}$	$\frac{S^2_i}{S^2_w}$																
Within	MEANS: <table><tr><td></td><td>Col A</td><td>Col B</td><td>Row Means</td></tr><tr><td>Row A</td><td>Cell</td><td>Cell</td><td>→</td></tr><tr><td>Row B</td><td>Cell</td><td>Cell</td><td>→</td></tr><tr><td>Column Means</td><td>↓</td><td>↓</td><td></td></tr></table>				Col A	Col B	Row Means	Row A	Cell	Cell	→	Row B	Cell	Cell	→	Column Means	↓	↓		X: individual score N: number of scores M _o : overall mean
	Col A	Col B	Row Means																	
Row A	Cell	Cell	→																	
Row B	Cell	Cell	→																	
Column Means	↓	↓																		
Total																				

Multivariate ANOVA (MANOVA)

≥ 2 DVs

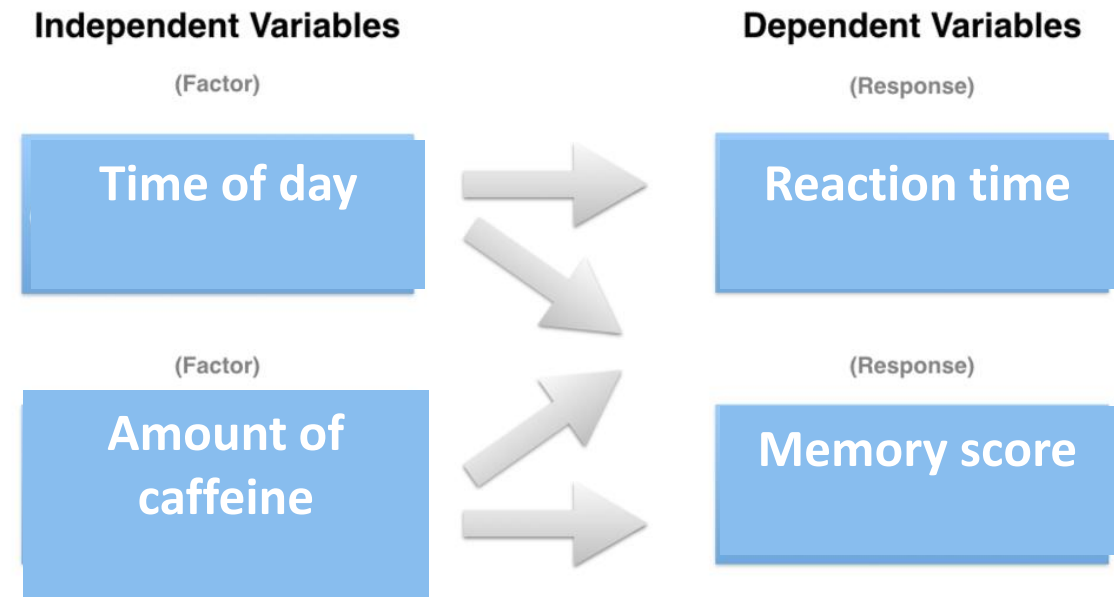
1 IV + 2 DVs

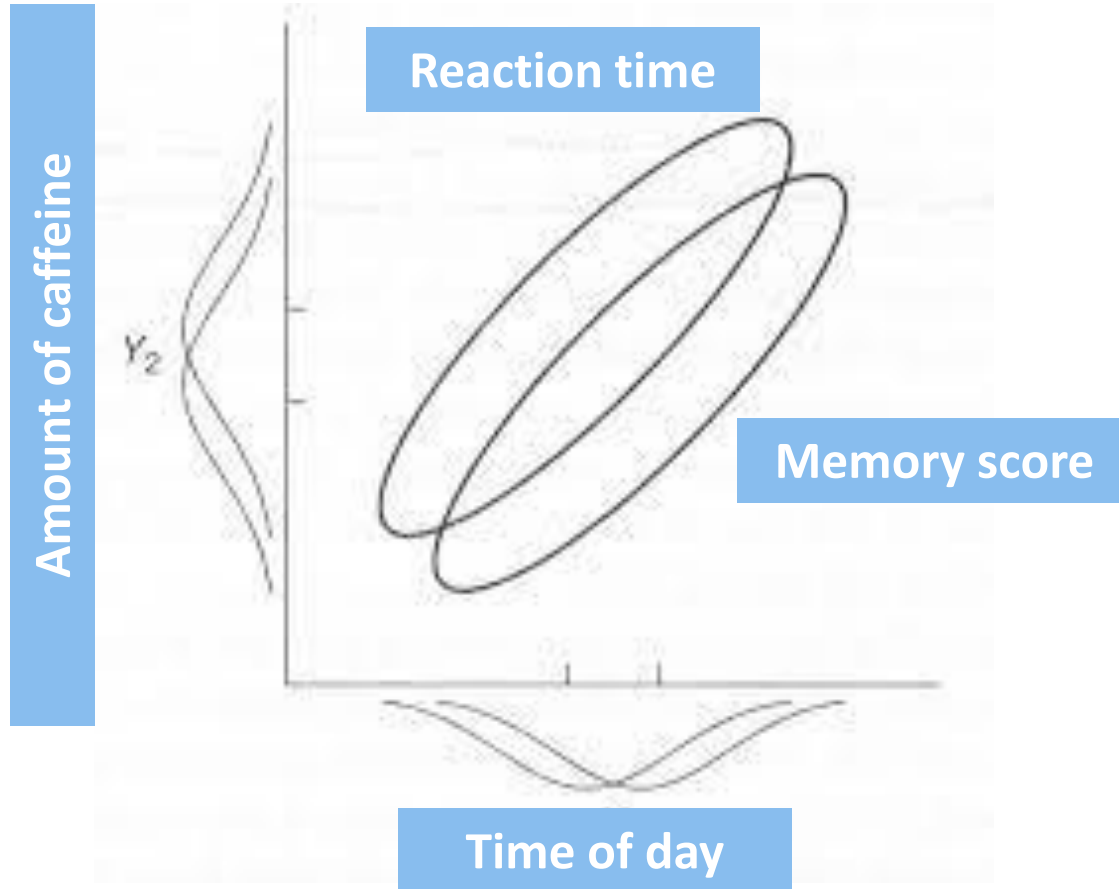
ONE-WAY MANOVA EXAMPLE



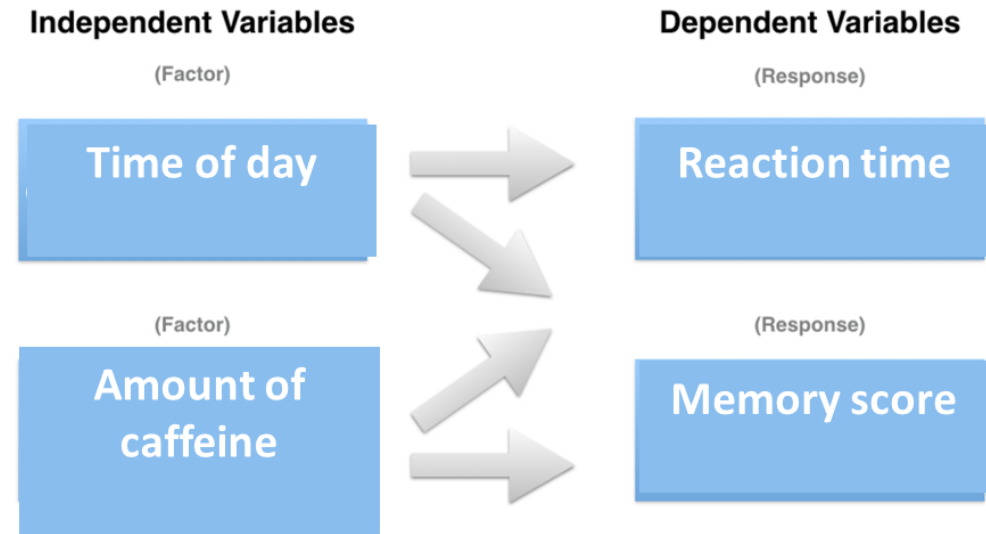
2 IVs + 2 DVs

TWO-WAY MANOVA EXAMPLE





TWO-WAY MANOVA EXAMPLE



Main effect - Time of Day?

Main effect - Amount of Caffeine?

Interaction? – MANOVA brings out interaction effects

linear combination of DVs to increase effects of IVs

Is MANOVA different from Repeated measures?

- Multiple DVs but not related
- They could be correlated/covariate

(Homogeneity of Covariances)

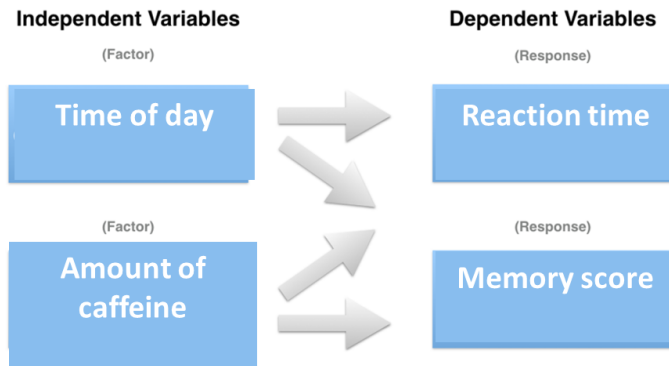
homogeneous variance → Pillai's trace test statistic

Non-homogenous variance → Wilks' lambda test statistic

Assumption of sphericity

- Same participant is repeated tested
- Relationship between multiple timepoints of DV
- Sphericity assumes these timepoints are homogenous
- Individual differences are removed

TWO-WAY MANOVA EXAMPLE



Reaction time → coffee → reaction time

		IV – categorical	DV – continuous (interval, ratio)
		Independent factor 1 IV > 2 groups	Dependent (Related) Samples 1 DV > 2 timepoints
Parametric (normal)	Homogeneous	One way ANOVA	Repeated measures ANOVA
	Non homogenous	Welch ANOVA	Sphericity correction
Non-parametric		<i>Kruskal-Wallis ANOVA</i>	Friedman's ANOVA

	>=2 IVs & 1 DV	>=2 IV, repeated DV	>=1 IV & >= 2 DVs
	Factorial ANOVA	Mixed ANOVA (>1 factors + Repeated measures)	MANOVA

IV= factor (these terms are used interchangeably)

Spot the difference

Younger vs older adults



Spot the difference

Younger vs older adults

Effect of distraction?



Spot the difference

Younger vs older adults

Effect of distraction?

IV 1 – age (young, old)

IV 2 – no distraction, unpleasant sounds

DV – no. of differences spotted (attention)



Spot the difference

Younger vs older adults

Effect of distraction?

Effect of reward?



Spot the difference

Younger vs older adults

Effect of distraction?

IV 1 – age (young, old)

IV 2 – no distraction, unpleasant sounds

DV – no. of differences spotted

Effect of reward?

IV 3– no reward, reward



Spot the difference

Younger vs older adults

Effect of distraction?

Can reward change attention levels in distractive environments?



Effect of distraction?

IV 1 – age (young, old)

IV 2 – no distraction, unpleasant sounds

DV – no. of differences spotted

Can reward change attention in
distractive environments?

Spot the difference

Younger vs older adults



Old

Young

Attention → reward → attention

distraction

no distraction

- Scenario 1: A fitness instructor wants to test the effectiveness of a performance-enhancing herbal supplement on students in his exercise class. The instructor gives Group A The herbal supplement and Group B receives the placebo. The students' fitness level is compared before and after six weeks of consuming the supplement or the sugar pill. His data do not support his hypothesis that the herbal supplement had an effect on fitness levels.

- Scenario 2: A social psychologist is interested discovering if whether women who are taller have a stronger career orientation. She measures height and gives a survey to women that measures many personality variables, including career orientation. Her data support her hypothesis that women who are taller had stronger career orientations.

Several weeks after Allen conducted a classroom experiment on the effectiveness of various metals in releasing hydrogen gas from hydrochloric acid, he read that the gas company was burying sheets of magnesium next to iron pipelines in order to prevent rusting. Allen wondered if other active metals would also be effective in preventing rust.

To investigate, he placed each of the following into a separate test tube containing water: one iron nail; one iron nail wrapped with an aluminum strip; one iron nail wrapped with a magnesium strip; and one iron nail wrapped in a lead strip. He used the same amounts of water from the same source, equal amounts of the metal wraps and the same type of iron nails. At the end of five days, he described the amounts of rusting either as small, moderate or large. He also recorded the color of the water.